Illustration 2

A man got a job whose initial monthly salary is fixed at ₹10,000 with an annual increment of ₹2000. His salary during 1st, 2nd and 3rd years will be ₹10000, ₹12000 and ₹14000 respectively.

If we now calculate the difference of the salaries for the successive years, we get $12000 - 10000 = 2000$; $14000 - 12000 = 2000$. Thus the difference between the successive numbers (salaries) is always 2000.

Did you observe the common property behind these two illustrations? In these two examples, the difference between successive terms always remains constant. Moreover, each term is obtained by adding a fixed number (2 and 2000 in illustrations 1 and 2 presented above) to the preceding term except the first term. This fixed number which is a constant for the differences between successive terms is called the "common difference".

Definition

Let a and d be real numbers. Then the numbers of the form a, $a+d$, $a+2d$, $a+3d$, $a+4d$, ... is said to form Arithmetic Progression denoted by A.P. The number 'a' is called the first term and 'd' is called the common difference.

Simply, an Arithmetic Progression is a sequence whose successive terms differ by a constant number. Thus, for example, the set of even positive integers 2, 4, 6, 8, 10, 12,... is an A.P. whose first term is $a = 2$ and common difference is also $d = 2$ since $4 - 2 = 2$, $6 - 4 = 2$, $8 - 6 = 2$, ...

Most of common real–life situations often produce numbers in A.P.

Note

- The difference between any two consecutive terms of an A.P. is always constant. That constant value is called the common difference.
- If there are finite numbers of terms in an A.P. then it is called Finite Arithmetic Progression. If there are infinitely many terms in an A.P. then it is called Infinite Arithmetic Progression.

2.7.1 Terms and Common Difference of an A.P.

1. The terms of an A.P. can be written as

$t_1 = a = a + (1-1)d$, $t_2 = a + d = a + (2-1)d$,

$t_3 = a + 2d = a + (3-1)d$, $t_4 = a + 3d = a + (4-1)d$, ...

In general, the nth term denoted by t_n can be written as $t_n = a + (n-1)d$.

In an AP, n^{th} term is, $t_n = a + (n-1)d$, here, a is the first term, d is the common difference.

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